ENGINEERING DECISION REPORT

Bifacial PV Mounting Decision Report

1) Veri Özeti Adapter Test Project için arka taşıyıcı gölgelenme etkisi karşılaştırılmış, referans senaryo 'No rear support' ve en iyi enerji sonucu 'No rear support' olarak belirlenmiştir.

Logo / Branding Area

PROJECT **Adapter Test Project**	REPORT DATE **2026-03-12**
REFERENCE SCENARIO **No rear support**	RECOMMENDED SCENARIO **No rear support**
SCENARIO COUNT **4**	SITE / COORDINATES **Bifacial Rear Support Comparison**

Report Metadata / Project Information

Context

Project Metadata

PREPARED BY Bifacial Analytics Platform	CLIENT Not specified
PROJECT CODE Not specified	REPORT VERSION 2.0
CURRENCY USD	ANNUAL ALBEDO N/A
ANNUAL GHI (KWH/M2) N/A	REFERENCE SCENARIO No rear support

Site Notes

Climate note not provided. NASA POWER and PVGIS context still applied where available.

Decision Context

This report is designed to support purlin-count selection by balancing energy yield, structural confidence, and indicative commercial implications.

BEST ENERGY SCENARIO

No rear support

Annual total: 1,108,000 kWh

BEST STRUCTURAL CONFIDENCE

3 rear purlins

Score: 84.5/100

LOWEST STRUCTURE COST

No rear support

Cost: +420,000 USD

Executive Summary

Investor / Management

1) Veri Özeti Adapter Test Project için arka taşıyıcı gölgelenme etkisi karşılaştırılmış, referans senaryo 'No rear support' ve en iyi enerji sonucu 'No rear support' olarak belirlenmiştir. Yıllık toplam enerji bandı 1,082,622 - 1,108,000 kWh olup senaryo yayılımı 25,378 kWh seviyesindedir. En iyi senaryonun referansa göre farkı +0 kWh (%+0.00) olarak hesaplanmıştır. 2) Teknik Yorum En iyi senaryoda arka blokaj %0.00, mismatch %10.00 ve etkili arka kazanç %10.80 düzeyindedir. Aşık/çapraz eleman yoğunluğu arttıkça mekanik rijitlik kazanımı ile arka yüz ışınım kaybı birlikte artabilir; nihai seçim enerji, montaj riski ve statik gereksinimlerin ortak optimizasyonu ile yapılmalıdır. 3) Tasarım Optimizasyon Önerileri Referans + 2 alternatif senaryo için çok kriterli karar matrisi kurulmalı; enerji, montaj zorluğu ve yapısal güvenlik birlikte puanlanmalıdır. Kritik eğim bölgelerinde segmentasyon (ör. 2x26 yerine 2x13) ile lokal kolon boyu ve sahadaki uygulama riski düşürülebilir. Uygulama öncesi, yüksek sertlik sınıfında ön delgi/kazık stratejisi ve ekipman erişimi için ayrı bir saha doğrulaması yapılmalıdır. 4) Riskler ve Kontrol Planı Sahada lokal topografya farklarının yüksek olduğu bölgeler için kolon boyu, bağlantı kotu ve erişilebilirlik kontrolleri ayrı hold-point olarak yönetilmelidir. Devreye alma öncesinde pilot bölgede ölçüm alınarak model kalibrasyonu yapılmalı ve gerekirse seçilen senaryo revize edilmelidir. 5) Sonuç Model sonuçları enerji açısından 'No rear support' senaryosunu öne çıkarmaktadır; nihai karar saha uygulanabilirliği ve statik emniyet ile birlikte verilmelidir.

Decision Snapshot

One-Page Decision

RECOMMENDED SCENARIO

No rear support

Decision score: 80.0/100

ENERGY DELTA VS REFERENCE

0 kWh

Positive values indicate higher annual total energy.

CAPEX DELTA VS REFERENCE

0 USD

Indicative structure CAPEX difference.

Why This Option

- Purlin count: 0
- No cross-brace
- Structure cost includes indicative purlin and brace uplift assumptions.

Watch-Outs WARNING

- No major downside is dominant in current model context; proceed with site-specific verification.

Primary recommendation: No rear support. Compared with reference 'No rear support', this option targets a balanced return profile under current climate/albedo assumptions.

Methodology Overview

Technical Basis

- Scenario set is evaluated with Monte Carlo-based rear-side obstruction and mismatch propagation.
- Rear gain and annual total energy are calculated per scenario and compared against reference.
- Decision score combines energy performance, indicative structure cost, and structural confidence proxy.

Data Sources and Assumptions

Input Integrity

SOURCE	PROVIDER	RETRIEVED	NOTE
NASA POWER	NASA	-	Albedo and climatology context.
PVGIS TMY 5.2	European Commission JRC	-	Irradiance and meteorological baseline.

ASSUMPTION	VALUE	CONFIDENCE
Analysis engine	Monte Carlo rear-side obstruction comparison across scenarios.	**HIGH**
Climate source	NASA POWER and PVGIS derived climate/albedo inputs.	**HIGH**
Cost normalization	Base structure cost = 420,000 USD, purlin uplift = 17,500 USD/purlin, cross-brace uplift = 12,500 USD.	**MEDIUM**
Uncertainty note	Ranges represent scenario confidence interpretation, not guaranteed yield bounds.	**MEDIUM**

Scenario Definitions

Alternatives

No rear support

- Purlin count: 0
- Cross-brace: No
- Rear blockage: 0.00%
- Structure cost: +420,000 USD

2 rear purlins

- Purlin count: 2
- Cross-brace: No
- Rear blockage: 6.50%
- Structure cost: +455,000 USD

3 rear purlins

- Purlin count: 3
- Cross-brace: No
- Rear blockage: 9.93%
- Structure cost: +472,500 USD

2 purlins + cross brace

- Purlin count: 2
- Cross-brace: Yes
- Rear blockage: 19.84%
- Structure cost: +467,500 USD

Scenario Comparison Summary

Side-by-Side

#	SCENARIO	PURLIN	CROSS BRACE	ANNUAL TOTAL (KWH)	EFF. REAR GAIN (%)	STRUCTURAL SCORE	STRUCTURE COST	DELTA VS REF (KWH)
1	**No rear support (REF)**	0	No	1,108,000	10.80	56.0	+420,000 USD	0
2	**2 rear purlins**	2	No	1,096,925	9.69	75.0	+455,000 USD	-11,075
3	**3 rear purlins**	3	No	1,093,217	9.32	84.5	+472,500 USD	-14,783
4	**2 purlins + cross brace**	2	Yes	1,082,622	8.26	82.5	+467,500 USD	-25,378

Bifacial Rear-Side Impact Analysis

Optical / Mismatch

Rear-Side Obstruction by Scenario

No rear support	0.00%
2 rear purlins	6.50%
3 rear purlins	9.93%
2 purlins + cross brace	19.84%

Higher values indicate stronger backside shading/obstruction effect.

Effective Rear Gain (%)

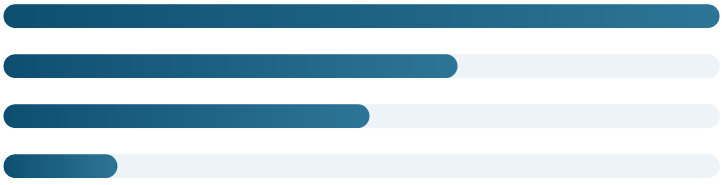
No rear support	10.80%
2 rear purlins	9.69%
3 rear purlins	9.32%
2 purlins + cross brace	8.26%

Estimated effective rear-side energy contribution ratio.

Energy Yield Comparison

Yield Delta

Annual Total Energy (kWh)



Comparative annual total energy output across purlin alternatives.

Structural / Engineering Considerations

Engineering

Structural Confidence Proxy (0-100)

No rear support	56.0
2 rear purlins	75.0
3 rear purlins	84.5
2 purlins + cross brace	82.5

Proxy score intended for comparative decision support, not final static approval.

Engineering Interpretation

- Structural confidence scores are comparative indicators based on purlin count and brace presence.
- Final structural acceptance requires project code checks, geotechnical envelope, and detailed design verification.
- Reference scenario for cost deltas: No rear support.

Cost and Commercial Comparison

Commercial

Indicative Structure Cost (USD)

No rear support	420,000
2 rear purlins	455,000
3 rear purlins	472,500
2 purlins + cross brace	467,500

Structure CAPEX proxy driven by purlin count and brace usage assumptions.

SCENARIO	STRUCTURE COST	DELTA VS REF	LCOE DELTA (%)
No rear support	+420,000 USD	0 USD	N/A
2 rear purlins	+455,000 USD	+35,000 USD	N/A
3 rear purlins	+472,500 USD	+52,500 USD	N/A
2 purlins + cross brace	+467,500 USD	+47,500 USD	N/A

Uncertainty / Sensitivity Interpretation

Risk Envelope

> Monte Carlo-based results are interpreted as confidence-aware scenario outcomes. Values indicate decision-oriented ranges, not deterministic guarantees.

SCENARIO	P90 (KWH)	P50 (KWH)	P10 (KWH)	INTERPRETATION
No rear support	1,108,000	1,108,000	1,108,000	Indicative confidence range derived from rear-side non-uniformity proxy.
2 rear purlins	1,092,435	1,096,925	1,101,415	Indicative confidence range derived from rear-side non-uniformity proxy.
3 rear purlins	1,088,739	1,093,217	1,097,696	Indicative confidence range derived from rear-side non-uniformity proxy.
2 purlins + cross brace	1,078,282	1,082,622	1,086,962	Indicative confidence range derived from rear-side non-uniformity proxy.

Key Trade-Offs

Decision Drivers

- Lower purlin count tends to improve rear irradiance access and can reduce structural CAPEX uplift.
- Higher purlin count increases structure robustness confidence but usually raises backside obstruction and cost.
- The economically preferred option should satisfy both yield advantage and minimum structural reliability threshold.

Final Recommendation

Decision

RECOMMENDED SCENARIO

No rear support

Recommended baseline for bankable deployment: 'No rear support'. Decision score 80.0/100.

- Energy-positioned outcome with annual total 1,108,000 kWh.
- Structure cost delta vs reference: +0 USD (indicative).
- Structural confidence proxy: 56.0/100.

If static checks or site risk controls tighten, '2 rear purlins' can be selected as a risk-downside alternative.

Investor Interpretation: Select the highest decision-score scenario while validating static constraints with project-specific engineering calculations.

Appendix / Technical Notes

Reference

- This report is for decision support and does not replace detailed structural design checks.
- Final EPC execution decisions should include local geotechnical constraints and code compliance.

SCENARIO ID	LABEL	PURLIN	CROSS BRACE	REAR BLOCKAGE (%)	MISMATCH (%)	DECISION SCORE
no-rear-support	No rear support	0	No	0.00	10.00	80.0
2-rear-purlins	2 rear purlins	2	No	6.50	13.62	51.5
3-rear-purlins	3 rear purlins	3	No	9.93	13.75	40.9
2-purlins-cross-brace	2 purlins + cross brace	2	Yes	19.84	14.10	21.5

Simulation Trace (Excerpt)

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No simulation trace provided.
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